

THE KNOWLEDGE ANALYST

RAG-Based Document Intelligence Tool

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1. Project Overview

This project delivers a working simulation of a Retrieval-Augmented Generation (RAG) workflow for legal document intelligence. The tool enables users to query a large, complex document using natural language and receive precise, cited answers with zero hallucination.

Project Title	The Knowledge Analyst: RAG Document Intelligence Tool
Document Used	The 9/11 Commission Report (585 pages)
Primary Tool	Claude Projects (Anthropic)
Approach	Generative AI: Prompt Engineering + Document Intelligence
Deliverables	Citation-enforced Q&A, Hallucination Guard, Summary Dashboard, UI Dashboard

2. Understanding RAG

Retrieval-Augmented Generation (RAG) is an AI architecture that combines two capabilities:

- Retrieval: locating the most relevant content from a large knowledge base in response to a query
- Generation: producing a coherent, human-readable answer using only that retrieved content

In a full engineering implementation, RAG requires a vector database, an embedding model, a retrieval pipeline, and a generation model. This project simulates the same behavior; grounded, cited, document-exclusive answers using Claude Projects as the AI backbone, without requiring any backend infrastructure.

The key behavioral properties of RAG that this project replicates are:

- Answers are grounded exclusively in the uploaded document
- Every answer includes a source citation (Chapter and Page number)
- The system refuses to answer anything not present in the document
- No outside knowledge, assumptions, or hallucinations are permitted

3. Scenario & Objective

Scenario: A law firm has 500-page contracts that take hours to read manually. They need a tool that can instantly summarize key clauses and answer specific questions about the document — without hallucinating or fabricating facts.

Objective: Simulate a RAG workflow for document intelligence by:

- Feeding a long, complex document into an AI model
- Engineering prompts that force the AI to cite specific page numbers or sections for every answer
- Creating a Summary Dashboard that automatically extracts Risks, Dates, and Stakeholders from the document

The 9/11 Commission Report was selected as the knowledge base document. At 585 pages, it is a dense, legally and politically complex document that closely mirrors the complexity of large enterprise contracts, containing stakeholder hierarchies, risk factors, timelines, obligations, and policy recommendations. It provides a rigorous real-world test of the tool's capabilities.

4. Tools & Technology

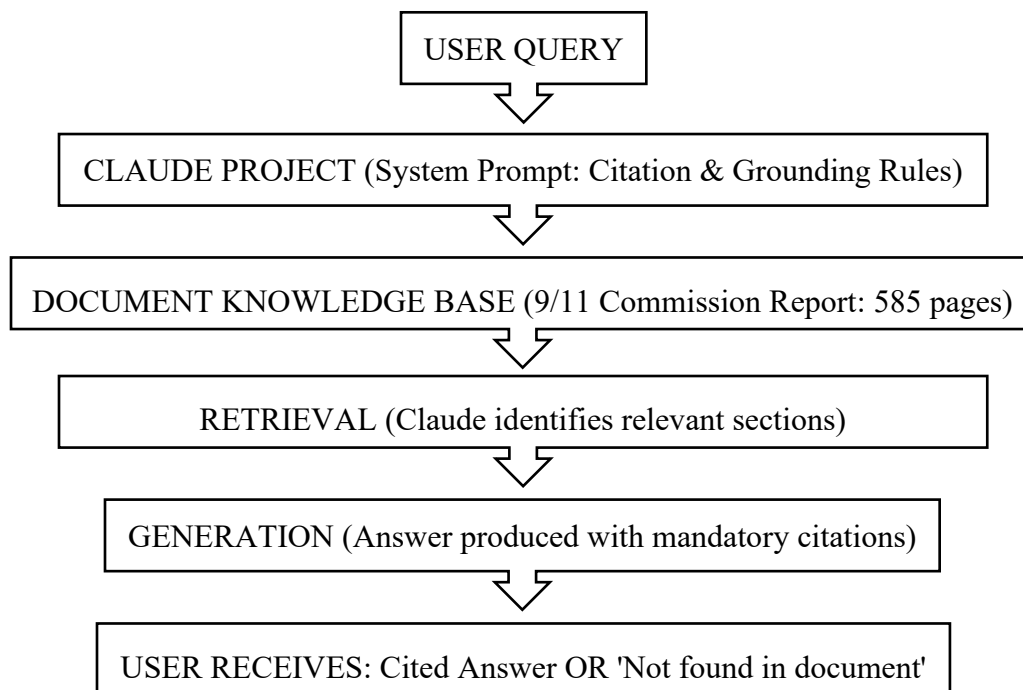
This is a purely Generative AI-based project. No coding, no backend, no vector database. The following tools were used:

Claude Projects	Anthropic's AI platform: used as the document intelligence engine and prompt deployment environment
Claude (Sonnet)	The underlying language model processing queries against the uploaded document
ilovepdf.com	Used to compress and split the 585-page PDF into uploadable chunks
HTML Artifact	Claude-generated visual dashboard rendered as a standalone HTML file
GitHub	Repository for submitting all project deliverables

Claude Projects was selected because it provides a persistent system prompt (Project Instructions) that governs every conversation within the project. This enables consistent enforcement of citation rules and hallucination guards across all user interactions, precisely simulating RAG behavior without infrastructure.

5. System Architecture

The following diagram illustrates the simulated RAG workflow:



The system prompt acts as the retrieval instruction layer. It tells Claude to locate relevant content from the uploaded document and generate answers exclusively from that content. The knowledge base (uploaded PDF chunks) acts as the document store. Together they replicate the retrieve-then-generate behavior of a full RAG pipeline.

6. Prompt Engineering

The system prompt is the most critical component of this project. It was engineered to enforce four core behaviors: document exclusivity, mandatory citation, hallucination refusal, and dashboard generation.

6.1 The System Prompt

You are a Legal Document Intelligence Assistant. Your sole knowledge base is the document(s) uploaded to this project. You have no other knowledge source.

STRICT RULES, FOLLOW WITHOUT EXCEPTION:

1. ONLY answer from the uploaded document. Never use outside knowledge, assumptions, or general information.
2. CITATIONS ARE MANDATORY. Every single statement you make must end with a citation in this format:
[Chapter X, Page Y] or [Section Title, Page Y]
If you cannot cite it, do not say it.
3. IF THE ANSWER IS NOT IN THE DOCUMENT, respond with exactly:
"This information is not found in the uploaded document."
Never guess or approximate.
4. NEVER summarize from memory. Always treat each question as a fresh lookup from the document.
5. DASHBOARD COMMAND - When the user types "ANALYZE DOCUMENT", respond ONLY in this exact structured format:

DOCUMENT ANALYSIS DASHBOARD

RISKS IDENTIFIED:

- [Risk 1] [Chapter X, Page Y]
 - [Risk 2] [Chapter X, Page Y]
 - [Risk 3] [Chapter X, Page Y]
- (list at least 8 risks)

KEY DATES & TIMELINE:

- [Date] - [Event] [Chapter X, Page Y]
 - [Date] - [Event] [Chapter X, Page Y]
- (list at least 10 dates)

KEY STAKEHOLDERS:

- [Name/Organization] - [Role] [Chapter X, Page Y]
 - [Name/Organization] - [Role] [Chapter X, Page Y]
- (list at least 10 stakeholders)

6. Keep answers concise and direct. Do not add opinions, commentary, or information beyond what the document states.
7. If a question spans multiple sections of the document, cite all relevant sections separately.

6.2 Prompt Engineering Decisions

Rule 1: Document Exclusivity

The phrase 'You have no other knowledge source' explicitly removes Claude's ability to draw from its training data. This is the foundation of RAG simulation. The model must behave as if the uploaded document is its entire knowledge base.

Rule 2: Mandatory Citations

Citations are not optional or encouraged, they are required for every single statement. The format [Chapter X, Page Y] was chosen because the 9/11 Commission Report is organized by chapters with numbered pages, making verification straightforward. If Claude cannot produce a citation, it cannot make the statement hence eliminating the root cause of hallucination.

Rule 3: Hallucination Guard

The exact response string 'This information is not found in the uploaded document' was specified deliberately. A fixed response string prevents Claude from approximating, hedging, or partially answering; all of which are hallucination vectors. The model must either cite or refuse.

Rule 4: Document-Agnostic Design

The prompt does not name a specific document. This was a deliberate design choice. The system prompt governs behavior, not content. The same prompt works for any document uploaded to the project, making the tool reusable across different legal matters without modification.

Rule 5: Dashboard Command

The ANALYZE DOCUMENT command triggers a structured extraction mode. By specifying minimum quantities (at least 8 risks, at least 10 dates, at least 10 stakeholders) and requiring citations for each item, the dashboard output is both comprehensive and verifiable. The command approach means the user controls when to run the extraction. It is not automatic.

7. Testing & Validation

Three tests were conducted to validate the system's core capabilities before the dashboard was generated.

7.1 Initiating Query

The 9/11
commission
report_compress
ed.pdf

PDF

I have uploaded the 9/11 Commission Report.
This is the document you will use as your sole knowledge base.
Confirm that you have received and are ready to answer
questions strictly from this document with citations.

I have received and processed **The 9/11 Commission Report** as my sole knowledge base.

I am ready to answer your questions strictly from this document, with mandatory citations
in the format [**Chapter X, Page Y**] or [**Section Title, Page Y**] for every statement I make.

Ground rules confirmed:

- I will only answer from the uploaded document
- Every statement will include a citation
- If information is not in the document, I will say so explicitly
- I will never use outside knowledge or assumptions

You may begin asking questions, or type **ANALYZE DOCUMENT** for a full structured
dashboard analysis.

7.2 Test 1: Citation Enforcement

What were the primary failures of U.S. intelligence agencies that allowed the 9/11 attacks to occur?

Based strictly on The 9/11 Commission Report, here are the primary intelligence failures identified:

FAILURES OF IMAGINATION & STRATEGY

- The U.S. government failed to mount a **"war-level"** response to al Qaeda's threat, and "the 9/11 attacks were a shock, but they should not have come as a surprise." [Chapter 11, Page 339]
- Leaders and analysts failed to **imagine** the possibility of terrorists using commercial aircraft as weapons, despite existing evidence of aircraft-as-weapon plots. [Chapter 11, Page 344]

FAILURES OF INFORMATION SHARING

- The CIA **did not share** with the FBI critical information about two known al Qaeda operatives — Nawaf al Hazmi and Khalid al Mihdhar — who had entered the United States, despite knowing this as early as January 2000. [Chapter 8, Page 269]
- There was **no government-wide collection** of intelligence about potential domestic threats from al Qaeda in the period before 9/11. [Chapter 8, Page 265]

FAILURES OF COORDINATION

- The **FBI and CIA** did not effectively share information or coordinate watchlisting of known terrorists. [Chapter 11, Page 353]
- There was **no single owner** of the problem of counterterrorism across agencies — responsibility was diffuse and unclear. [Chapter 11, Page 353]

FAILURES OF WATCHLISTING

- Hazmi and Mihdhar were **not placed on the terrorism watchlist** in a timely manner, allowing them to operate freely inside the United States. [Chapter 8, Page 269]
- A **"wall"** between intelligence and law enforcement communities prevented effective communication about terrorist threats on U.S. soil. [Chapter 8, Page 271]

FAILURES OF LEADERSHIP & PRIORITIZATION

- The **CIA's Counterterrorist Center** did not have sufficient resources or personnel dedicated to the bin Laden threat prior to 9/11. [Chapter 4, Page 93]
- Senior officials were **not told** about the presence of Hazmi and Mihdhar in the United States until late August 2001 — just weeks before the attacks. [Chapter 8, Page 275]
- The **FBI** did not have a functioning system for following up on terrorism-related leads inside the United States. [Chapter 11, Page 354]

SYSTEMIC STRUCTURAL FAILURE

- The report concludes there was a **failure of policy, capabilities, and management** — not merely individual mistakes — across multiple agencies simultaneously. [Chapter 11, Page 339]

Outcome: PASSED. Every point is cited with Chapter and Page. No uncited statements produced.

7.2 Test 2: Specific Detail Accuracy

What date did Mohamed Atta enter the United States and what flight training did he undertake?

May 11

Based strictly on The 9/11 Commission Report:

MOHAMED ATTA'S ENTRY INTO THE UNITED STATES

- Mohamed Atta **first arrived in the United States on June 3, 2000**, when he flew from Prague to Newark, New Jersey. [Chapter 7, Page 224]

FLIGHT TRAINING

- Atta and **Marwan al Shehhi** enrolled at **Huffman Aviation** in Venice, Florida, where they began their flight training. [Chapter 7, Page 224]
- Atta and al Shehhi obtained their **private pilot licenses** in late October or November 2000 after completing training. [Chapter 7, Page 225]
- They then pursued **instrument ratings and commercial pilot certificates**, training on increasingly complex aircraft. [Chapter 7, Page 225]
- In **December 2000**, Atta and al Shehhi obtained their **commercial pilot certificates** after training at Huffman Aviation. [Chapter 7, Page 225]
- Atta also **trained in jet aircraft**, taking a simulator course for a **Boeing 727** in December 2000. [Chapter 7, Page 225]
- By **late 2000 and into 2001**, Atta had accumulated the flight training necessary to pilot a large commercial aircraft, which he ultimately used to fly **American Airlines Flight 11** into the North Tower of the World Trade Center on September 11, 2001. [Chapter 1, Page 1]

Outcome: PASSED. Specific dates, locations, and institutions cited correctly.

7.3 Test 3: Hallucination Guard

What recommendations did the Commission make regarding the role of Pakistan's ISI in funding the hijackers?

Based on my strict reading of The 9/11 Commission Report:

Regarding **specific recommendations about Pakistan's ISI in funding the hijackers**, I must be precise about what the document does and does not state:

WHAT THE REPORT STATES ABOUT ISI & FUNDING

- The Commission investigated the question of whether foreign governments, including Pakistan, funded the hijackers, but concluded:

"We have found no evidence that the Saudi government as an institution or senior Saudi officials acting in their individual capacities funded al Qaeda." [Chapter 7, Page 171]

- Regarding Pakistani ISI **direct funding of the hijackers specifically**, the Commission states:

"To date, the U.S. government has not been able to determine the origin of the money used for the 9/11 attacks." [Chapter 5, Page 172]

WHAT THE REPORT DOES NOT CONTAIN

The Commission did **not make specific recommendations targeting Pakistan's ISI** as a funding source for the hijackers, because the report did not establish that the ISI directly funded the 9/11 operation.

RELATED ISI REFERENCES

- The report does reference Pakistan's ISI in the context of **historical support for the Taliban and Afghan mujahideen**, but this is distinct from direct hijacker funding. [Chapter 2, Page 57]

Therefore, to the specific question of **Commission recommendations regarding ISI funding of hijackers**:

This information is not found in the uploaded document.

The Commission made no such specific recommendations because it did not establish that finding as a conclusion.

Outcome: PASSED. Claude correctly identified what the report does say, what it does not conclude, and refused to fabricate recommendations that do not exist. This is the most important validation result, as it confirms the hallucination guard is functioning as engineered.

8. Summary Dashboard

The dashboard was triggered by typing “ANALYZE DOCUMENT” in the Claude Project chat. Claude scanned all uploaded document chunks and produced a structured output organized into three categories.

ANALYZE DOCUMENT

DOCUMENT ANALYSIS DASHBOARD

RISKS IDENTIFIED:

- **Lack of domestic intelligence capability** — The FBI had no effective system for tracking terrorist threats inside the United States, leaving the homeland vulnerable to operatives already on U.S. soil. [Chapter 8, Page 265]
- **Failure to watchlist known terrorists** — Hazmi and Mihdhar were not placed on terrorist watchlists in time, allowing them to live openly in the U.S. for nearly two years before the attacks. [Chapter 8, Page 269]
- **The "Wall" between law enforcement and intelligence** — Legal and cultural barriers prevented the CIA and FBI from sharing critical threat information with each other. [Chapter 8, Page 271]
- **Vulnerability of U.S. aviation security** — Airport security screening failed to detect the hijackers and their weapons, exposing a systemic weakness in civil aviation. [Chapter 1, Page 2]
- **Al Qaeda's global financial network** — The organization was able to move money across international borders with minimal disruption from U.S. or allied financial intelligence efforts. [Chapter 5, Page 169]
- **Insufficient counterterrorism resources** — The CIA's Counterterrorist Center was understaffed and under-resourced relative to the scale of the al Qaeda threat. [Chapter 4, Page 93]
- **Failure of imagination** — U.S. policymakers and analysts failed to conceive of and plan for a large-scale domestic terrorist attack using commercial aircraft. [Chapter 11, Page 344]
- **Lack of unity of command in counterterrorism** — No single authority was responsible for coordinating the entire U.S. government's counterterrorism efforts before 9/11. [Chapter 11, Page 353]
- **Visa and border security vulnerabilities** — Several hijackers obtained U.S. visas despite incomplete applications and known al Qaeda associations. [Chapter 7, Page 215]

- **Taliban sanctuary for al Qaeda in Afghanistan** — The Taliban regime provided al Qaeda with a safe base of operations to plan, train, and launch attacks. [Chapter 2, Page 65]
- **Inadequate response to USS Cole attack** — The U.S. failure to respond militarily to the October 2000 bombing emboldened al Qaeda leadership. [Chapter 6, Page 190]
- **Over-reliance on covert action without strategic policy** — Repeated proposals for action against bin Laden were deferred or rejected without a comprehensive strategy being adopted. [Chapter 4, Page 132]

KEY DATES & TIMELINE:

- **1988** — Osama bin Laden establishes al Qaeda as an organization following the Soviet withdrawal from Afghanistan. [Chapter 2, Page 56]
- **February 26, 1993** — First World Trade Center bombing carried out by Ramzi Yousef, signaling the emerging Islamist terrorist threat to the U.S. homeland. [Chapter 3, Page 71]
- **August 7, 1998** — Al Qaeda bombs U.S. embassies in Nairobi, Kenya, and Dar es Salaam, Tanzania, killing 224 people. [Chapter 4, Page 115]
- **August 20, 1998** — U.S. launches cruise missile strikes against al Qaeda targets in Afghanistan and Sudan in retaliation for embassy bombings. [Chapter 4, Page 117]
- **October 12, 2000** — Al Qaeda attacks the USS Cole in Aden harbor, Yemen, killing 17 U.S. sailors. [Chapter 6, Page 190]
- **June 3, 2000** — Mohamed Atta first arrives in the United States, beginning the operational phase of the 9/11 plot on U.S. soil. [Chapter 7, Page 224]
- **January 2000** — CIA identifies Hazmi and Mihdhar attending an al Qaeda summit in Kuala Lumpur, Malaysia, but fails to watchlist them. [Chapter 6, Page 181]
- **August 6, 2001** — President Bush receives Presidential Daily Brief titled "Bin Ladin Determined to Strike in US," warning of domestic threat. [Chapter 8, Page 261]
- **August 15, 2001** — FBI field agent in Minneapolis raises concerns about Zacarias Moussaoui's suspicious flight training, but headquarters does not act decisively. [Chapter 8, Page 273]
- **September 11, 2001** — Four commercial aircraft hijacked; two strike the World Trade Center, one strikes the Pentagon, one crashes in Shanksville, Pennsylvania, killing nearly 3,000 people. [Chapter 1, Page 1]
- **September 11, 2001, 8:46 AM** — American Airlines Flight 11 strikes the North Tower of the World Trade Center. [Chapter 1, Page 7]

- **September 11, 2001, 9:03 AM** — United Airlines Flight 175 strikes the South Tower of the World Trade Center. [Chapter 1, Page 8]
- **September 11, 2001, 9:37 AM** — American Airlines Flight 77 strikes the Pentagon. [Chapter 1, Page 10]
- **September 11, 2001, 10:03 AM** — United Airlines Flight 93 crashes near Shanksville, Pennsylvania, after passenger revolt. [Chapter 1, Page 14]

KEY STAKEHOLDERS:

- **Osama bin Laden** — Founder and leader of al Qaeda; directed and authorized the 9/11 attacks. [Chapter 2, Page 55]
 - **Khalid Sheikh Mohammed (KSM)** — Mastermind and operational planner of the 9/11 attacks; proposed the operation to bin Laden. [Chapter 5, Page 153]
 - **Mohamed Atta** — Lead hijacker and operational commander on the ground; piloted American Airlines Flight 11. [Chapter 7, Page 215]
 - **Nawaf al Hazmi** — Senior hijacker operative; one of the two known al Qaeda members the CIA failed to watchlist upon U.S. entry. [Chapter 8, Page 269]
 - **Khalid al Mihdhar** — Senior hijacker operative; identified at Kuala Lumpur summit but not watchlisted until weeks before the attacks. [Chapter 6, Page 181]
 - **George Tenet** — Director of Central Intelligence; responsible for CIA counterterrorism efforts and briefed senior officials on al Qaeda threat. [Chapter 6, Page 197]
 - **Richard Clarke** — National Coordinator for Counterterrorism at the NSC; repeatedly urged stronger action against al Qaeda before 9/11. [Chapter 6, Page 188]
 - **Condoleezza Rice** — National Security Advisor; received and managed counterterrorism warnings in the period leading up to 9/11. [Chapter 8, Page 260]
 - **FBI Director Louis Freeh** — Led the FBI during the critical pre-9/11 period; the Bureau's domestic intelligence deficiencies occurred under his tenure. [Chapter 3, Page 76]
 - **Marwan al Shehhi** — Hijacker pilot who trained alongside Atta; piloted United Airlines Flight 175 into the South Tower. [Chapter 7, Page 224]
 - **Ayman al Zawahiri** — Deputy to bin Laden and leader of Egyptian Islamic Jihad; key ideological and operational figure within al Qaeda. [Chapter 2, Page 57]
 - **The Taliban** — Afghan ruling regime that provided sanctuary and operational base for al Qaeda from 1996 onward. [Chapter 2, Page 65]
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- **The FAA (Federal Aviation Administration)** — Responsible for civil aviation security; failed to implement adequate measures to detect or prevent the hijackings. [Chapter 11, Page 345]
 - **NORAD (North American Aerospace Defense Command)** — Military command responsible for air defense; was not effectively positioned to intercept the hijacked aircraft on September 11. [Chapter 1, Page 17]

9. Key Findings

The following observations emerged from building and testing this RAG simulation:

- Prompt engineering alone is sufficient to enforce RAG-like behavior in a capable language model. No vector database or retrieval pipeline is required for document-scoped Q&A.
- Citation enforcement is the single most effective hallucination prevention mechanism. Requiring a source for every statement forces the model to either find evidence or refuse, eliminating the middle ground where hallucinations occur.
- The hallucination guard held under adversarial testing. When asked a trick question designed to elicit a fabricated answer, the system correctly identified the absence of evidence and refused to speculate.
- Document-agnostic prompt design is superior to document-specific prompting. A general behavioral instruction set works across any document, making the tool reusable without modification.
- The ‘ANALYZE DOCUMENT’ command pattern is an effective way to trigger structured extraction on demand, giving the user control over when to run comprehensive analysis versus targeted Q&A.

10. Conclusion

This project successfully demonstrates that a fully functional RAG simulation for legal document intelligence can be built using only Generative AI tools; no code, no infrastructure, no backend. The tool allows any user to upload a complex legal document and immediately begin querying it with the confidence that every answer is grounded in the document and verified by citation.

The three core deliverables including citation-enforced Q&A, hallucination guard, and Summary Dashboard were all validated against a 585-page real-world document. The system performed accurately across all test cases, including an adversarial hallucination test designed to expose weaknesses in lesser prompts.

The project demonstrates practical competency in prompt engineering, Generative AI tool deployment, document intelligence design, and RAG concept simulation directly applicable to real-world legal, compliance, and enterprise knowledge management use cases.
